2 Data Import and Wrangling

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1 Chapter 2: Data Import and Data Wrangling

1.1 Agenda

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- **Part 3 (30 mins)** – Handling Missing Data
- **Part 4 (20 mins)** – Practice and Q&A

1.2 Part 1: Reading Different Types of Data (30 mins)

We can also check our working directory using the command getwd().

1.2.1 CSV File Example: Iris Dataset

```r
# We can import csv file using the base library using the command read.csv
iris_data <- read.csv("../data/Iris.csv")

# Or we can use readr library using the command read_csv to import csv
library(readr)
iris_data_r <- read_csv("../data/Iris.csv")
```

```
Rows: 150 Columns: 6
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): Species
dbl (5): Id, SepalLengthCm, SepalWidthCm, PetalLengthCm, PetalWidthCm

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# View column names
names(iris_data)
```

```
[1] "Id"            "SepalLengthCm" "SepalWidthCm"  "PetalLengthCm"
[5] "PetalWidthCm"  "Species"
```

```r
# View first few rows
head(iris_data)
```

```
  Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm     Species
1  1           5.1          3.5           1.4          0.2 Iris-setosa
2  2           4.9          3.0           1.4          0.2 Iris-setosa
3  3           4.7          3.2           1.3          0.2 Iris-setosa
4  4           4.6          3.1           1.5          0.2 Iris-setosa
5  5           5.0          3.6           1.4          0.2 Iris-setosa
6  6           5.4          3.9           1.7          0.4 Iris-setosa
```

```r
# Check structure
str(iris_data)
```

```
'data.frame':	150 obs. of  6 variables:
 $ Id           : int  1 2 3 4 5 6 7 8 9 10 ...
 $ SepalLengthCm: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ SepalWidthCm : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ PetalLengthCm: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ PetalWidthCm : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species      : chr  "Iris-setosa" "Iris-setosa" "Iris-setosa" "Iris-
setosa" ...
```

1.2.2 Understanding the Difference: read.csv() vs read_csv()

```r
# Base R: read.csv()
class(iris_data)        # Returns "data.frame"
```

```
[1] "data.frame"
```

```r
# readr: read_csv()
class(iris_data_r)      # Returns "spec_tbl_df" "tbl_df" "tbl" "data.frame"
```

```
[1] "spec_tbl_df" "tbl_df"      "tbl"         "data.frame"
```

```r
# Both work, but read_csv() is generally faster and has better defaults
```

1.2.3 Quick Data Exploration

```r
# Dimensions
dim(iris_data)
```

```
[1] 150   6
```

```r
# Summary statistics
summary(iris_data)
```

```
       Id         SepalLengthCm    SepalWidthCm    PetalLengthCm  
 Min.   :  1.00   Min.   :4.300   Min.   :2.000   Min.   :1.000  
 1st Qu.: 38.25   1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600  
 Median : 75.50   Median :5.800   Median :3.000   Median :4.350  
 Mean   : 75.50   Mean   :5.843   Mean   :3.054   Mean   :3.759  
 3rd Qu.:112.75   3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100  
 Max.   :150.00   Max.   :7.900   Max.   :4.400   Max.   :6.900  
  PetalWidthCm     Species         
 Min.   :0.100   Length:150        
 1st Qu.:0.300   Class :character  
 Median :1.300   Mode  :character  
 Mean   :1.199                     
 3rd Qu.:1.800                     
 Max.   :2.500                     
```

```r
# View in RStudio
# View(iris_data)
```

1.3 Part 2: Data Cleaning and Manipulation using `dplyr` (40 mins)

1.3.1 Load Required Packages

```r
library(dplyr)
```

```

Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

1.3.2 The Pipe Operator: %>%

The pipe operator `%>%` makes code more readable by chaining operations together.

```r
# Without pipe (hard to read)
head(filter(iris_data, SepalLengthCm > 6))
```

```
  Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm         Species
1 51           7.0          3.2           4.7          1.4 Iris-versicolor
2 52           6.4          3.2           4.5          1.5 Iris-versicolor
3 53           6.9          3.1           4.9          1.5 Iris-versicolor
4 55           6.5          2.8           4.6          1.5 Iris-versicolor
5 57           6.3          3.3           4.7          1.6 Iris-versicolor
6 59           6.6          2.9           4.6          1.3 Iris-versicolor
```

```r
# With pipe (easier to read)
iris_data %>%
  filter(SepalLengthCm > 6) %>%
  head()
```

```
  Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm         Species
1 51           7.0          3.2           4.7          1.4 Iris-versicolor
2 52           6.4          3.2           4.5          1.5 Iris-versicolor
3 53           6.9          3.1           4.9          1.5 Iris-versicolor
4 55           6.5          2.8           4.6          1.5 Iris-versicolor
5 57           6.3          3.3           4.7          1.6 Iris-versicolor
6 59           6.6          2.9           4.6          1.3 Iris-versicolor
```

Keyboard shortcut: Ctrl + Shift + M (Windows) or Cmd + Shift + M (Mac)

1.3.3 Basic Wrangling Examples

1.3.3.1 Selecting Columns with `select()`

```
# Select specific columns
iris_data %>%
  select(Species, SepalLengthCm, PetalLengthCm) %>%
  head()
```

```
      Species SepalLengthCm PetalLengthCm
1 Iris-setosa           5.1           1.4
2 Iris-setosa           4.9           1.4
3 Iris-setosa           4.7           1.3
4 Iris-setosa           4.6           1.5
5 Iris-setosa           5.0           1.4
6 Iris-setosa           5.4           1.7
```

```
# Select columns using patterns
iris_data %>%
  select(starts_with("Petal")) %>%
  head()
```

```
  PetalLengthCm PetalWidthCm
1           1.4          0.2
2           1.4          0.2
3           1.3          0.2
4           1.5          0.2
5           1.4          0.2
6           1.7          0.4
```

```
iris_data %>%
  select(ends_with("Cm")) %>%
  head()
```

```
  SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm
1           5.1          3.5           1.4          0.2
2           4.9          3.0           1.4          0.2
3           4.7          3.2           1.3          0.2
4           4.6          3.1           1.5          0.2
5           5.0          3.6           1.4          0.2
6           5.4          3.9           1.7          0.4
```

```r
# Exclude columns with minus sign
iris_data %>%
  select(-Id) %>%
  head()
```

```
  SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm     Species
1           5.1          3.5           1.4          0.2 Iris-setosa
2           4.9          3.0           1.4          0.2 Iris-setosa
3           4.7          3.2           1.3          0.2 Iris-setosa
4           4.6          3.1           1.5          0.2 Iris-setosa
5           5.0          3.6           1.4          0.2 Iris-setosa
6           5.4          3.9           1.7          0.4 Iris-setosa
```

1.3.3.2 Filtering Rows with `filter()`

```r
# Filter for setosa species only
iris_data %>%
  filter(Species == "Iris-setosa") %>%
  head()
```

```
  Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm     Species
1  1           5.1          3.5           1.4          0.2 Iris-setosa
2  2           4.9          3.0           1.4          0.2 Iris-setosa
3  3           4.7          3.2           1.3          0.2 Iris-setosa
4  4           4.6          3.1           1.5          0.2 Iris-setosa
5  5           5.0          3.6           1.4          0.2 Iris-setosa
6  6           5.4          3.9           1.7          0.4 Iris-setosa
```

```r
# Filter flowers with long petals
iris_data %>%
  filter(PetalLengthCm > 5) %>%
  head()
```

```
   Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm         Species
1  84           6.0          2.7           5.1          1.6 Iris-versicolor
2 101           6.3          3.3           6.0          2.5  Iris-virginica
3 102           5.8          2.7           5.1          1.9  Iris-virginica
4 103           7.1          3.0           5.9          2.1  Iris-virginica
5 104           6.3          2.9           5.6          1.8  Iris-virginica
6 105           6.5          3.0           5.8          2.2  Iris-virginica
```

```r
# Multiple conditions with AND (&)
iris_data %>%
  filter(Species == "Iris-virginica" & SepalLengthCm > 6.5) %>%
  head()
```

```
   Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm        Species
1 103           7.1          3.0           5.9          2.1 Iris-virginica
2 106           7.6          3.0           6.6          2.1 Iris-virginica
3 108           7.3          2.9           6.3          1.8 Iris-virginica
4 109           6.7          2.5           5.8          1.8 Iris-virginica
5 110           7.2          3.6           6.1          2.5 Iris-virginica
6 113           6.8          3.0           5.5          2.1 Iris-virginica
```

```r
# Multiple conditions with OR (|)
iris_data %>%
  filter(PetalLengthCm > 6 | PetalWidthCm > 2) %>%
  head()
```

```
   Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm        Species
1 101           6.3          3.3           6.0          2.5 Iris-virginica
2 103           7.1          3.0           5.9          2.1 Iris-virginica
3 105           6.5          3.0           5.8          2.2 Iris-virginica
4 106           7.6          3.0           6.6          2.1 Iris-virginica
5 108           7.3          2.9           6.3          1.8 Iris-virginica
6 110           7.2          3.6           6.1          2.5 Iris-virginica
```

```r
# Using %in% operator for multiple values
iris_data %>%
  filter(Species %in% c("Iris-setosa", "Iris-versicolor")) %>%
  head()
```

```
  Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm     Species
1  1           5.1          3.5           1.4          0.2 Iris-setosa
2  2           4.9          3.0           1.4          0.2 Iris-setosa
3  3           4.7          3.2           1.3          0.2 Iris-setosa
4  4           4.6          3.1           1.5          0.2 Iris-setosa
5  5           5.0          3.6           1.4          0.2 Iris-setosa
6  6           5.4          3.9           1.7          0.4 Iris-setosa
```

1.3.3.3 Creating New Variables with `mutate()`

```
# Calculate areas
iris_data <- iris_data %>%
  mutate(
    SepalArea = SepalLengthCm * SepalWidthCm,
    PetalArea = PetalLengthCm * PetalWidthCm
  )

head(iris_data)
```

```
  Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm     Species
1  1           5.1          3.5           1.4          0.2 Iris-setosa
2  2           4.9          3.0           1.4          0.2 Iris-setosa
3  3           4.7          3.2           1.3          0.2 Iris-setosa
4  4           4.6          3.1           1.5          0.2 Iris-setosa
5  5           5.0          3.6           1.4          0.2 Iris-setosa
6  6           5.4          3.9           1.7          0.4 Iris-setosa
  SepalArea PetalArea
1     17.85      0.28
2     14.70      0.28
3     15.04      0.26
4     14.26      0.30
5     18.00      0.28
6     21.06      0.68
```

```
# Create categorical variables
iris_data <- iris_data %>%
  mutate(
    SizeCategory = ifelse(PetalLengthCm > 4, "Large", "Small"),
    SepalRatio = SepalLengthCm / SepalWidthCm
  )

head(iris_data)
```

```
  Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm     Species
1  1           5.1          3.5           1.4          0.2 Iris-setosa
2  2           4.9          3.0           1.4          0.2 Iris-setosa
3  3           4.7          3.2           1.3          0.2 Iris-setosa
4  4           4.6          3.1           1.5          0.2 Iris-setosa
5  5           5.0          3.6           1.4          0.2 Iris-setosa
```

```
6  6           5.4          3.9           1.7          0.4 Iris-setosa
  SepalArea PetalArea SizeCategory SepalRatio
1     17.85      0.28        Small   1.457143
2     14.70      0.28        Small   1.633333
3     15.04      0.26        Small   1.468750
4     14.26      0.30        Small   1.483871
5     18.00      0.28        Small   1.388889
6     21.06      0.68        Small   1.384615
```

```r
# Using case_when() for multiple conditions
iris_data <- iris_data %>%
  mutate(
    PetalSize = case_when(
      PetalLengthCm < 2 ~ "Small",
      PetalLengthCm < 5 ~ "Medium",
      TRUE ~ "Large"
    )
  )

head(iris_data)
```

```
  Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm     Species
1  1           5.1          3.5           1.4          0.2 Iris-setosa
2  2           4.9          3.0           1.4          0.2 Iris-setosa
3  3           4.7          3.2           1.3          0.2 Iris-setosa
4  4           4.6          3.1           1.5          0.2 Iris-setosa
5  5           5.0          3.6           1.4          0.2 Iris-setosa
6  6           5.4          3.9           1.7          0.4 Iris-setosa
  SepalArea PetalArea SizeCategory SepalRatio PetalSize
1     17.85      0.28        Small   1.457143     Small
2     14.70      0.28        Small   1.633333     Small
3     15.04      0.26        Small   1.468750     Small
4     14.26      0.30        Small   1.483871     Small
5     18.00      0.28        Small   1.388889     Small
6     21.06      0.68        Small   1.384615     Small
```

1.3.3.4 Sorting Rows with `arrange()`

```r
# Sort by Petal Length (ascending)
iris_data %>%
  arrange(PetalLengthCm) %>%
  head()
```

```
  Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm     Species
1 23           4.6          3.6           1.0          0.2 Iris-setosa
2 14           4.3          3.0           1.1          0.1 Iris-setosa
3 15           5.8          4.0           1.2          0.2 Iris-setosa
4 36           5.0          3.2           1.2          0.2 Iris-setosa
5  3           4.7          3.2           1.3          0.2 Iris-setosa
6 17           5.4          3.9           1.3          0.4 Iris-setosa
  SepalArea PetalArea SizeCategory SepalRatio PetalSize
1     16.56      0.20        Small   1.277778     Small
2     12.90      0.11        Small   1.433333     Small
3     23.20      0.24        Small   1.450000     Small
4     16.00      0.24        Small   1.562500     Small
5     15.04      0.26        Small   1.468750     Small
6     21.06      0.52        Small   1.384615     Small
```

```r
# Sort by Petal Length (descending)
iris_data %>%
  arrange(desc(PetalLengthCm)) %>%
  head()
```

```
   Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm        Species
1 119           7.7          2.6           6.9          2.3 Iris-virginica
2 118           7.7          3.8           6.7          2.2 Iris-virginica
3 123           7.7          2.8           6.7          2.0 Iris-virginica
4 106           7.6          3.0           6.6          2.1 Iris-virginica
5 132           7.9          3.8           6.4          2.0 Iris-virginica
6 108           7.3          2.9           6.3          1.8 Iris-virginica
  SepalArea PetalArea SizeCategory SepalRatio PetalSize
1     20.02     15.87        Large   2.961538     Large
2     29.26     14.74        Large   2.026316     Large
3     21.56     13.40        Large   2.750000     Large
4     22.80     13.86        Large   2.533333     Large
5     30.02     12.80        Large   2.078947     Large
6     21.17     11.34        Large   2.517241     Large
```

```r
# Sort by multiple columns
iris_data %>%
  arrange(Species, desc(SepalLengthCm)) %>%
  head()
```

```
  Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm     Species
```

```
1 15          5.8         4.0          1.2          0.2 Iris-setosa
2 16          5.7         4.4          1.5          0.4 Iris-setosa
3 19          5.7         3.8          1.7          0.3 Iris-setosa
4 34          5.5         4.2          1.4          0.2 Iris-setosa
5 37          5.5         3.5          1.3          0.2 Iris-setosa
6  6          5.4         3.9          1.7          0.4 Iris-setosa
  SepalArea PetalArea SizeCategory SepalRatio PetalSize
1     23.20      0.24        Small   1.450000     Small
2     25.08      0.60        Small   1.295455     Small
3     21.66      0.51        Small   1.500000     Small
4     23.10      0.28        Small   1.309524     Small
5     19.25      0.26        Small   1.571429     Small
6     21.06      0.68        Small   1.384615     Small
```

1.3.3.5 Summarizing with `group_by()` and `summarize()`

```
# Overall summary
iris_data %>%
  summarize(
    avg_sepal_length = mean(SepalLengthCm, na.rm = TRUE),
    avg_petal_length = mean(PetalLengthCm, na.rm = TRUE),
    total_flowers = n()
  )
```

```
  avg_sepal_length avg_petal_length total_flowers
1         5.843333         3.758667           150
```

```
# Summary by species
iris_data %>%
  group_by(Species) %>%
  summarize(
    avg_sepal_length = mean(SepalLengthCm, na.rm = TRUE),
    avg_sepal_width = mean(SepalWidthCm, na.rm = TRUE),
    avg_petal_length = mean(PetalLengthCm, na.rm = TRUE),
    avg_petal_width = mean(PetalWidthCm, na.rm = TRUE),
    count = n()
  )
```

```
# A tibble: 3 x 6
  Species      avg_sepal_length avg_sepal_width avg_petal_length avg_petal_width
  <chr>                   <dbl>           <dbl>            <dbl>           <dbl>
```

```
1 Iris-setosa              5.01           3.42             1.46           0.244
2 Iris-versic~             5.94           2.77             4.26           1.33
3 Iris-virgin~             6.59           2.97             5.55           2.03
# i 1 more variable: count <int>
```

```
# Summary by multiple groups
iris_data %>%
  group_by(Species, SizeCategory) %>%
  summarize(
    avg_sepal_length = mean(SepalLengthCm, na.rm = TRUE),
    count = n(),
    .groups = 'drop'
  )
```

```
# A tibble: 4 x 4
  Species         SizeCategory avg_sepal_length count
  <chr>           <chr>                   <dbl> <int>
1 Iris-setosa     Small                    5.01    50
2 Iris-versicolor Large                    6.15    34
3 Iris-versicolor Small                    5.49    16
4 Iris-virginica  Large                    6.59    50
```

1.3.4 Combining Multiple Operations

```
# Complex pipeline: Find large virginica flowers and their stats
iris_data %>%
  filter(Species == "Iris-virginica", PetalLengthCm > 6) %>%
  select(Species, SepalLengthCm, PetalLengthCm, SepalArea, PetalArea) %>%
  arrange(desc(PetalArea)) %>%
  head(10)
```

```
         Species SepalLengthCm PetalLengthCm SepalArea PetalArea
1 Iris-virginica           7.7           6.9     20.02     15.87
2 Iris-virginica           7.2           6.1     25.92     15.25
3 Iris-virginica           7.7           6.7     29.26     14.74
4 Iris-virginica           7.7           6.1     23.10     14.03
5 Iris-virginica           7.6           6.6     22.80     13.86
6 Iris-virginica           7.7           6.7     21.56     13.40
7 Iris-virginica           7.9           6.4     30.02     12.80
8 Iris-virginica           7.4           6.1     20.72     11.59
9 Iris-virginica           7.3           6.3     21.17     11.34
```

1.3.5 Additional Useful Functions

1.3.5.1 Count rows with `count()`

```r
# Count flowers by species
iris_data %>%
  count(Species)
```

```
          Species  n
1     Iris-setosa 50
2 Iris-versicolor 50
3  Iris-virginica 50
```

```r
# Count with sorting
iris_data %>%
  count(Species, sort = TRUE)
```

```
          Species  n
1     Iris-setosa 50
2 Iris-versicolor 50
3  Iris-virginica 50
```

```r
# Count by multiple variables
iris_data %>%
  count(Species, SizeCategory)
```

```
          Species SizeCategory  n
1     Iris-setosa        Small 50
2 Iris-versicolor        Large 34
3 Iris-versicolor        Small 16
4  Iris-virginica        Large 50
```

1.3.5.2 Get unique values with `distinct()`

```r
# Get unique species
iris_data %>%
  distinct(Species)
```

```
          Species
1     Iris-setosa
2 Iris-versicolor
3  Iris-virginica
```

```r
# Get unique combinations
iris_data %>%
  distinct(Species, PetalSize)
```

```
          Species PetalSize
1     Iris-setosa     Small
2 Iris-versicolor    Medium
3 Iris-versicolor     Large
4  Iris-virginica     Large
5  Iris-virginica    Medium
```

1.4 Part 3: Handling Missing Data (30 mins)

1.4.1 Checking Missing Values

```r
# Load naniar for missing data visualization
library(naniar)

# Check for missing values
miss_var_summary(iris_data)
```

```
# A tibble: 11 x 3
   variable      n_miss pct_miss
   <chr>          <int>    <num>
 1 Id                 0        0
 2 SepalLengthCm      0        0
 3 SepalWidthCm       0        0
 4 PetalLengthCm      0        0
 5 PetalWidthCm       0        0
 6 Species            0        0
 7 SepalArea          0        0
 8 PetalArea          0        0
```

```
 9 SizeCategory       0        0
10 SepalRatio         0        0
11 PetalSize          0        0
```

```r
# Check if any values are missing
any(is.na(iris_data))
```

```
[1] FALSE
```

```r
# Count NAs in each column
colSums(is.na(iris_data))
```

```
           Id SepalLengthCm  SepalWidthCm PetalLengthCm  PetalWidthCm
            0             0             0             0             0
      Species     SepalArea     PetalArea  SizeCategory    SepalRatio
            0             0             0             0             0
    PetalSize
            0
```

1.4.2 Creating Sample Data with Missing Values

For demonstration, let's introduce some missing values:

```r
# Create a copy with missing values
iris_missing <- iris_data
set.seed(123)  # For reproducibility

# Randomly introduce NAs
iris_missing$SepalLengthCm[sample(1:nrow(iris_missing), 10)] <- NA
iris_missing$PetalWidthCm[sample(1:nrow(iris_missing), 15)] <- NA

# Check missing data
miss_var_summary(iris_missing)
```

```
# A tibble: 11 x 3
   variable      n_miss pct_miss
   <chr>          <int>    <num>
 1 PetalWidthCm      15    10
 2 SepalLengthCm     10     6.67
 3 Id                 0     0
```

```
 4 SepalWidthCm       0      0
 5 PetalLengthCm      0      0
 6 Species            0      0
 7 SepalArea          0      0
 8 PetalArea          0      0
 9 SizeCategory       0      0
10 SepalRatio         0      0
11 PetalSize          0      0
```

1.4.3 Removing Missing Values

```r
# Remove rows with any missing values
iris_complete <- na.omit(iris_missing)

cat("Original rows:", nrow(iris_missing), "\n")
```

```
Original rows: 150
```

```r
cat("After removing NAs:", nrow(iris_complete), "\n")
```

```
After removing NAs: 127
```

```r
# Remove only if specific column has NA
iris_filtered <- iris_missing %>%
  filter(!is.na(SepalLengthCm))

cat("After filtering SepalLengthCm:", nrow(iris_filtered), "\n")
```

```
After filtering SepalLengthCm: 140
```

1.4.4 Replacing Missing Values

```r
# Replace NAs with mean
iris_mean <- iris_missing %>%
  mutate(
    SepalLengthCm = ifelse(is.na(SepalLengthCm),
                           mean(SepalLengthCm, na.rm = TRUE),
```

```
                          SepalLengthCm),
    PetalWidthCm = ifelse(is.na(PetalWidthCm),
                         mean(PetalWidthCm, na.rm = TRUE),
                         PetalWidthCm)
  )

# Verify
miss_var_summary(iris_mean)
```

```
# A tibble: 11 x 3
   variable      n_miss pct_miss
   <chr>          <int>    <num>
 1 Id                 0        0
 2 SepalLengthCm      0        0
 3 SepalWidthCm       0        0
 4 PetalLengthCm      0        0
 5 PetalWidthCm       0        0
 6 Species            0        0
 7 SepalArea          0        0
 8 PetalArea          0        0
 9 SizeCategory       0        0
10 SepalRatio         0        0
11 PetalSize          0        0
```

```
# Replace with median (more robust to outliers)
iris_median <- iris_missing %>%
  mutate(
    SepalLengthCm = ifelse(is.na(SepalLengthCm),
                          median(SepalLengthCm, na.rm = TRUE),
                          SepalLengthCm),
    PetalWidthCm = ifelse(is.na(PetalWidthCm),
                         median(PetalWidthCm, na.rm = TRUE),
                         PetalWidthCm)
  )

# Replace with species-specific mean (group imputation)
iris_group_mean <- iris_missing %>%
  group_by(Species) %>%
  mutate(
    SepalLengthCm = ifelse(is.na(SepalLengthCm),
                          mean(SepalLengthCm, na.rm = TRUE),
```

```
                              SepalLengthCm),
    PetalWidthCm = ifelse(is.na(PetalWidthCm),
                             mean(PetalWidthCm, na.rm = TRUE),
                             PetalWidthCm)
  ) %>%
  ungroup()

miss_var_summary(iris_group_mean)
```

```
# A tibble: 11 x 3
   variable      n_miss pct_miss
   <chr>          <int>    <num>
 1 Id                 0        0
 2 SepalLengthCm      0        0
 3 SepalWidthCm       0        0
 4 PetalLengthCm      0        0
 5 PetalWidthCm       0        0
 6 Species            0        0
 7 SepalArea          0        0
 8 PetalArea          0        0
 9 SizeCategory       0        0
10 SepalRatio         0        0
11 PetalSize          0        0
```

1.4.5 Handling NAs During Analysis

```
# Without na.rm - returns NA
mean(iris_missing$SepalLengthCm)
```

```
[1] NA
```

```
# With na.rm = TRUE - ignores NAs
mean(iris_missing$SepalLengthCm, na.rm = TRUE)
```

```
[1] 5.855714
```

```r
# Summary with NA handling
iris_missing %>%
  summarize(
    mean_sepal = mean(SepalLengthCm, na.rm = TRUE),
    median_sepal = median(SepalLengthCm, na.rm = TRUE),
    count_non_na = sum(!is.na(SepalLengthCm)),
    count_na = sum(is.na(SepalLengthCm))
  )
```

```
  mean_sepal median_sepal count_non_na count_na
1   5.855714          5.8          140       10
```

1.5 Part 4: Practice and Q&A (20 mins)

1.5.1 Practice Tasks (based on iris data)

1.5.1.1 Task 1: Import and Explore

- Import the iris dataset
- Display the first 10 rows
- Check the number of rows and columns

```r
# Solution
iris_practice <- read.csv("../data/Iris.csv")
head(iris_practice, 10)
```

```
   Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm     Species
1   1           5.1          3.5           1.4          0.2 Iris-setosa
2   2           4.9          3.0           1.4          0.2 Iris-setosa
3   3           4.7          3.2           1.3          0.2 Iris-setosa
4   4           4.6          3.1           1.5          0.2 Iris-setosa
5   5           5.0          3.6           1.4          0.2 Iris-setosa
6   6           5.4          3.9           1.7          0.4 Iris-setosa
7   7           4.6          3.4           1.4          0.3 Iris-setosa
8   8           5.0          3.4           1.5          0.2 Iris-setosa
9   9           4.4          2.9           1.4          0.2 Iris-setosa
10 10           4.9          3.1           1.5          0.1 Iris-setosa
```

```
dim(iris_practice)
```

```
[1] 150   6
```

1.5.1.2 Task 2: Filter and Select

- Select only Iris-virginica flowers with PetalLengthCm > 6
- Show only Species, SepalLengthCm, and PetalLengthCm columns

```
# Solution
iris_practice %>%
  filter(Species == "Iris-virginica", PetalLengthCm > 6) %>%
  select(Species, SepalLengthCm, PetalLengthCm)
```

```
         Species SepalLengthCm PetalLengthCm
1 Iris-virginica           7.6           6.6
2 Iris-virginica           7.3           6.3
3 Iris-virginica           7.2           6.1
4 Iris-virginica           7.7           6.7
5 Iris-virginica           7.7           6.9
6 Iris-virginica           7.7           6.7
7 Iris-virginica           7.4           6.1
8 Iris-virginica           7.9           6.4
9 Iris-virginica           7.7           6.1
```

1.5.1.3 Task 3: Create New Variables

- Create a variable for PetalRatio = PetalLengthCm / PetalWidthCm
- Create a variable for SepalRatio = SepalLengthCm / SepalWidthCm

```
# Solution
iris_practice <- iris_practice %>%
  mutate(
    PetalRatio = PetalLengthCm / PetalWidthCm,
    SepalRatio = SepalLengthCm / SepalWidthCm
  )

head(iris_practice)
```

```
  Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm     Species
1  1           5.1          3.5           1.4          0.2 Iris-setosa
2  2           4.9          3.0           1.4          0.2 Iris-setosa
3  3           4.7          3.2           1.3          0.2 Iris-setosa
4  4           4.6          3.1           1.5          0.2 Iris-setosa
5  5           5.0          3.6           1.4          0.2 Iris-setosa
6  6           5.4          3.9           1.7          0.4 Iris-setosa
  PetalRatio SepalRatio
1       7.00   1.457143
2       7.00   1.633333
3       6.50   1.468750
4       7.50   1.483871
5       7.00   1.388889
6       4.25   1.384615
```

1.5.1.4 Task 4: Group and Summarize

- Group by Species
- Calculate average SepalLengthCm, PetalLengthCm, and count per species

```
# Solution
iris_practice %>%
  group_by(Species) %>%
  summarize(
    avg_sepal_length = mean(SepalLengthCm, na.rm = TRUE),
    avg_petal_length = mean(PetalLengthCm, na.rm = TRUE),
    count = n()
  )
```

```
# A tibble: 3 x 4
  Species         avg_sepal_length avg_petal_length count
  <chr>                      <dbl>            <dbl> <int>
1 Iris-setosa                 5.01             1.46    50
2 Iris-versicolor             5.94             4.26    50
3 Iris-virginica              6.59             5.55    50
```

1.5.1.5 Task 5: Complex Pipeline

- Filter for flowers with SepalLengthCm > 6
- Create a new variable TotalLength = SepalLengthCm + PetalLengthCm
- Group by Species

- Calculate average TotalLength
- Sort by average TotalLength descending

```
# Solution
iris_practice %>%
  filter(SepalLengthCm > 6) %>%
  mutate(TotalLength = SepalLengthCm + PetalLengthCm) %>%
  group_by(Species) %>%
  summarize(
    avg_total_length = mean(TotalLength, na.rm = TRUE),
    count = n()
  ) %>%
  arrange(desc(avg_total_length))
```

```
# A tibble: 2 x 3
  Species         avg_total_length count
  <chr>                      <dbl> <int>
1 Iris-virginica              12.5    41
2 Iris-versicolor             11.0    20
```

1.6 Further Resources (for Data Import & Wrangling)

- Tidyverse Documentation: https://www.tidyverse.org/packages/
- Importing Data in R: https://r4ds.hadley.nz/data-import.html
- Data Transformation Cheatsheet: https://posit.co/resources/cheatsheets/